Excellent Books

- *Turbulence and Transition Modeling*. Edited by M. Hallbäck, D.S. Henningson, A.V. Johansson, and P. H. Alfredsson. Kluwer, 1996.
- *Turbulence Modeling for CFD*. D. C. Wilcox. DCW Industries, 1998.

Algebraic Models

- B. Baldwin and H. Lomax. *Thin-layer approximation and algebraic model for separated turbulent flows.* AIAA conference paper, 1978. Also probably a NASA technical memorandum (TM) about that time. Algebraic model.
- D.A. Johnson and L.S. King. *A mathematically simple turbulence closure model for attached and separated turbulent boundary layers.* AIAA Journal, v 23 (11) pp 1684–1692, 1985. Algebraic model.

Baldwin-Barth (one-equation model)

- B. Baldwin and T. Barth. *A One-Equation Turbulence Transport Model for High Reynolds Number Wall-Bounded Flows.* AIAA conference paper, 1991. Also probably a NASA technical memorandum (TM) about that time.

Spalart-Allmaras (one-equation model)

- P. R. Spalart and S. R. Allmaras. *A one-equation turbulence model for aerodynamic flows.* Recherche Aerospatiale, v 1, pp 5–21, 1994 . Also an AIAA conference paper from about that time.

$k-\varepsilon$ Model

- W. P. Jones and B. E. Launder. *The Prediction of Laminarization with a Two-Equation Model of Turbulence.* Int. J. Heat and Mass Transfer, v. 15, pp 301–314, 1972. An early $k-\varepsilon$ paper.
- B. E. Launder and D. B. Spalding. *The numerical computation of turbulent flows.* Computer Methods in Applied Mechanics and Engineering, v 3 pp 269–289, 1974. An early $k-\varepsilon$ paper.

$k-\omega$ Model

- D.C. Wilcox. *Reassessment of the scale-determining equation for advanced turbulence models.* AIAA Journal, v 26 (11) pp 1299–1310, 1988. Not the first paper on the topic, but a reasonably good one.

- F.R. Menter. *Two-equation eddy-viscosity turbulence models for engineering applications.* AIAA Journal, v 32 (8) pp 1598–1605, 1994. This is the original paper on the $k-\omega$ SST model.

Algebraic stress models

- W. Rodi. *A new algebraic relation for calculating Reynolds stresses.* ZAMM, v 56, pp. 219ff.
- Or begin with Section 6.2 of Wilcox's book.

Reynolds stress models

- B. E. Launder, G. J. Reece, and W Rodi. *Progress in development of a Reynolds-stress turbulence closure.* Journal of Fluid Mechanics, v 68, pp 537–566, 1975.

Large Eddy Simulation

- W. Rodi, J. H. Ferziger, M. Breuer, et al. *Status of large eddy simulation: Results of a workshop.* Journal of Fluids Engineering — Transactions of the ASME, v 119 (2), pp 248–262, June, 1997.
- J. H. Ferziger. *Recent advances in Large-Eddy Simulation.* Engineering Turbulence Modeling and Experiments 3, (W. Rodi and G. Begeles, eds.) Elsevier, 1996.
- W. Rodi. *Large-Eddy Simulations of the flow past bluff bodies — State-of-the-art.* Japan Society of Mechanical Engineers International Journal, Series B, v 41 (2).

Wall-Modeled LES

- J. Larsson, S. Kawai, J. Bodart, and I. Bermejo-Moreno. *Large eddy simulation with modeled wall-stress: recent progress and future directions.* Mechanical Engineering Reviews, v 3 (1), 2016.
- S. T. Bose and G. I. Park. *Wall-Modeled Large-Eddy Simulation for Complex Turbulent Flows.* Annual Review of Fluid Mechanics, v 50, pp 535–561, 2018.

Direct Numerical Simulation

- P. Moin and K. Mahesh. *Direct numerical simulation: A tool in turbulence research.* Annual Review Of Fluid Mechanics, v 30, pp 539–578, 1998.